

OLS Regression - Diagnostic Audit Report

Target: home_value | Predictors: price, sq_footage, year_built, garage_area, lot_size, bedrooms, bathrooms, school_rating, crime_index, commute_min, walkability_score | Best Model: Linear | n = 150

Correlation Heatmap — All Predictors & Target



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Diagnostic Scorecard - Best Model (Linear)

Test	Statistic	p-value	Threshold	Result	Notes
F-Test (Overall)	493.5393	0.0000	p < 0.05	PASS	df model=11, resid=138
Normality - Shapiro-Wilk	0.9879	0.2185	p > 0.05	PASS	Tests residual normality
Normality - Jarque-Bera	2.5468	0.2799	p > 0.05	PASS	Skew=0.262, Kurt=2.636
Homoscedasticity - Breusch-Pagan	8.9612	0.6255	p > 0.05	PASS	Tests constant variance
Autocorrelation - Durbin-Watson	1.9428	-	1.5 – 2.5	PASS	Tests residual independence
Multicollinearity - Max VIF	1.15	-	< 5	PASS	Predictors: price, sq_footage, year_built, garage_area, lot_size, bedrooms, bathrooms, school_rating, crime_index, commute_min, walkability_score
Influential Obs - Cook's D	7 flagged	-	4/n = 0.0267	FAIL	Red circles on residual plot

OLS Regression - Interpretation & Model Comparison

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Interpretation of Results

Overall Fit	The best model (degree 1) explains 97.5% of variance in home_value ($R^2 = 0.9752$, Adj $R^2 = 0.9732$), indicating a strong fit.
Model Significance (F-Test)	$F(11, 138) = 493.539$, $p = 0.0000$. The overall regression is statistically significant ($p < 0.05$).
Normality of Residuals (Shapiro-Wilk)	$W = 0.9879$, $p = 0.2185$. Residuals appear normally distributed — the normality assumption is met.
Homoscedasticity (Breusch-Pagan)	BP stat = 8.9612, $p = 0.6255$. Residual variance appears constant (homoscedastic).
Autocorrelation (Durbin-Watson)	DW = 1.9428. No significant autocorrelation detected.
Multicollinearity (VIF)	Max VIF = 1.15. No multicollinearity concern (all VIF < 5).
Influential Observations (Cook's Distance)	7 influential observation(s) detected (Cook's D > 0.0267). Review flagged points for data entry errors or genuine outliers.

Top 5 Models Ranked by BIC

Models ranked by original-scale BIC — lower = better fit penalised for complexity. SW p = Shapiro-Wilk (normality); BP p = Breusch-Pagan (homoscedasticity). Green = PASS ($p > 0.05$), Red = FAIL ($p \leq 0.05$). Single-predictor runs also test Logarithmic, Exponential, Log-Log, Reciprocal, Spline, and Lag-1 variants, producing up to 10 candidates. Multi-predictor runs fit Linear and Interaction models only, so fewer rows appear.

Rank	Model Type	Fitted Equation	R ²	Adj R ²	AIC	BIC	SW p	BP p
1	Linear	home_value = 24.474 + 0.8762*price - 0.7809*sq_footage + 0.2766*year_built - 2.5309*garage_area + 0.6726*lot_size + 1.9835*bedrooms + 0.4221*bathrooms + 1.9294*school_rating + 1.1300*crime_index - 1.8439*commute_min + 2.0065*walkability_score	0.9752	0.9732	1417.04	1453.17	0.2185	0.6255
2	Polynomial (deg 2)	home_value = 24.474 + 0.8762*price - 0.7809*sq_footage + 0.2766*year_built - 2.5309*garage_area + 0.6726*lot_size + 1.9835*bedrooms + 0.4221*bathrooms + 1.9294*school_rating + 1.1300*crime_index - 1.8439*commute_min + 2.0065*walkability_score	0.9752	0.9732	1417.04	1453.17	0.2185	0.6255
3	Polynomial (deg 3)	home_value = 24.474 + 0.8762*price - 0.7809*sq_footage + 0.2766*year_built - 2.5309*garage_area + 0.6726*lot_size + 1.9835*bedrooms + 0.4221*bathrooms + 1.9294*school_rating + 1.1300*crime_index - 1.8439*commute_min + 2.0065*walkability_score	0.9752	0.9732	1417.04	1453.17	0.2185	0.6255
4	Polynomial (deg 4)	home_value = 24.474 + 0.8762*price - 0.7809*sq_footage + 0.2766*year_built - 2.5309*garage_area + 0.6726*lot_size + 1.9835*bedrooms + 0.4221*bathrooms + 1.9294*school_rating + 1.1300*crime_index - 1.8439*commute_min + 2.0065*walkability_score	0.9752	0.9732	1417.04	1453.17	0.2185	0.6255
5	Interaction	home_value = -8.215 + 0.3025*price + 0.1126*sq_footage + 0.3142*year_built - 3.0451*garage_area + 0.7703*lot_size + 1.6743*bedrooms + 0.5884*bathrooms + 1.4428*school_rating - 1.8090*crime_index - 0.7923*commute_min + 2.3596*walkability_score + 0.0039*price*sq_footage + 0.0114*price*year_built - 0.0028*price*garage_area - 0.0050*price*lot_size + 0.0019*price*bedrooms + 0.0025*price*bathrooms + 0.0017*price*school_rating + 0.0014*price*crime_index - 0.0021*price*commute_min - 0.0012*price*walkability_score - 0.0037*sq_footage*year_built - 0.0005*sq_footage*garage_area + 0.0037*sq_footage*lot_size + 0.0002*sq_footage*bedrooms - 0.0055*sq_footage*bathrooms - 0.0057*sq_footage*school_rating + 0.0021*sq_footage*crime_index - 0.0046*sq_footage*commute_min - 0.0060*sq_footage*walkability_score + 0.0039*year_built*garage_area - 0.0020*year_built*lot_size - 0.0012*year_built*bedrooms + 0.0061*year_built*bathrooms + 0.0018*year_built*school_rating - 0.0065*year_built*crime_index - 0.0109*year_built*commute_min + 0.0019*year_built*walkability_score - 0.0046*garage_area*lot_size + 0.0024*garage_area*bedrooms + 0.0038*garage_area*bathrooms - 0.0012*garage_area*school_rating - 0.0005*garage_area*crime_index + 0.0063*garage_area*commute_min + 0.0037*garage_area*walkability_score + 0.0010*lot_size*bedrooms + 0.0011*lot_size*bathrooms + 0.0037*lot_size*school_rating + 0.0032*lot_size*crime_index - 0.0019*lot_size*commute_min - 0.0023*lot_size*walkability_score + 0.0036*bedrooms*bathrooms + 0.0050*bedrooms*school_rating - 0.0025*bedrooms*crime_index - 0.0057*bedrooms*commute_min + 0.0024*bedrooms*walkability_score - 0.0020*bathrooms*school_rating - 0.0060*bathrooms*crime_index - 0.0029*bathrooms*commute_min - 0.0020*bathrooms*walkability_score - 0.0003*school_rating*crime_index + 0.0023*school_rating*commute_min + 0.0041*school_rating*walkability_score + 0.0025*crime_index*commute_min - 0.0040*crime_index*walkability_score - 0.0048*commute_min*walkability_score	0.9861	0.9751	1439.74	1641.45	0.43	0.9977

Diagnostic Plots - Best Model

OLS Diagnostic Plots

